

Quantitative Bitcoin Trading Strategy

Strategy Report & Backtest Summary

Project	quant-btc-strategy-backtester
Author	Arth Sh.
Program	Summer of Quant 2026 - Quant Community, IIT Bombay
Asset / Data	Bitcoin (BTC), Daily OHLCV, 2019-2023
Repository	github.com/ArthSh/quant-btc-strategy-backtester

A. Strategy Logic & Hypothesis

The core hypothesis behind this project is that Bitcoin, being a highly volatile and sentiment-driven asset, tends to move in persistent directional trends once a clear range breakout occurs, rather than reverting immediately to a mean. Discretionary trading decisions are difficult to make consistent and are hard to evaluate objectively, so the project instead tests this hypothesis systematically: a rule-based **Donchian Channel Breakout** system is used to detect genuine trend initiation, an EMA(50)/EMA(200) crossover filter is layered on top to confirm that the breakout aligns with the prevailing macro-trend (avoiding false breakouts inside choppy, range-bound regimes), and an ATR-based volatility filter and trailing stop are used to size risk relative to current market conditions and to lock in gains as a trend matures. The strategy trades both directions (long and short), which fits a 24/7, no-shorting-restriction market like BTC.

Entry / Exit Rules

Signal	Long	Short
Entry trigger	Price breaks above 25-day Donchian High	Price breaks below 25-day Donchian Low
Trend filter	EMA(50) > EMA(200)	EMA(50) < EMA(200)
Volatility filter	ATR-based confirmation	ATR-based confirmation
Exit	Exit Donchian channel breach or ATR trailing stop	Exit Donchian channel breach or ATR trailing stop

The signal library also includes a broader feature set (RSI, MACD, ROC, Momentum, Bollinger Bands, rolling volatility, ADX, relative volume, OBV, log/daily returns, candle-body and wick features) built during exploratory analysis, of which the trend and volatility features above were selected as the final, most decisive signals for this strategy.

B. Key Performance Metrics

The finalized, optimized configuration was backtested on daily BTC OHLCV data (2019-2023), evaluating look-ahead-bias-free, sequentially validated trade signals. Headline results:

Metric	Value	Interpretation
Sharpe Ratio	1.06	Positive risk-adjusted return over the test period

Net Profit	3,731.94	Cumulative profit in backtest currency units
Maximum Drawdown	46.43%	Largest peak-to-trough equity decline
Total Trades	20	Low trade frequency, consistent with trend-following

The framework additionally tracks a wider set of diagnostics for every configuration it evaluates – win rate, gross profit, average profit per trade, largest single win/loss, average drawdown, and holding-period statistics – giving a fuller risk picture beyond the four headline numbers above.

C. Parameter Optimization & Ranking

Rather than hand-picking parameters, an exhaustive grid search was run over three parameters – the ATR trailing-stop multiplier, the Donchian entry window, and the Donchian exit window – producing **196 distinct strategy configurations**. Each configuration was backtested independently and ranked on a composite of risk-adjusted and robustness criteria:

- Sharpe Ratio – primary ranking criterion (risk-adjusted return)
- Net Profit – absolute profitability
- Maximum Drawdown – downside risk / capital preservation
- Win Rate – consistency of individual trades
- Number of Trades – statistical significance / robustness of the sample

The configuration reported in Section B was the top-ranked candidate out of the 196 tested, selected for combining the highest risk-adjusted return (Sharpe 1.06) with an acceptable trade count (20) for statistical relevance, rather than for having the single highest net profit or win rate in isolation – reflecting a preference for robust, repeatable edge over curve-fit outperformance on any one metric.

Only one final strategy family (Donchian Breakout with EMA trend filter and ATR risk management) is submitted for this project; the ranking above reflects an internal comparison across its 196 parameter variants rather than a comparison across unrelated strategy types.

Methodology Notes

- Feature engineering across trend, momentum, volatility, trend-strength and volume indicators.
- Sequential (walk-forward-style) signal validation to eliminate look-ahead bias.
- Exhaustive grid search for parameter selection, avoiding manual/discretionary tuning.
- Final configuration chosen using risk-adjusted performance rather than raw profit alone.

Future Improvements

- Walk-forward optimization and multi-asset portfolio support
- Volatility-targeted position sizing
- Transaction cost / slippage modeling and Bayesian parameter search
- ML-based signal generation and live exchange API integration